SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

Beginner

Vectors in 3D

Cross Product

Equations of Lines and Planes

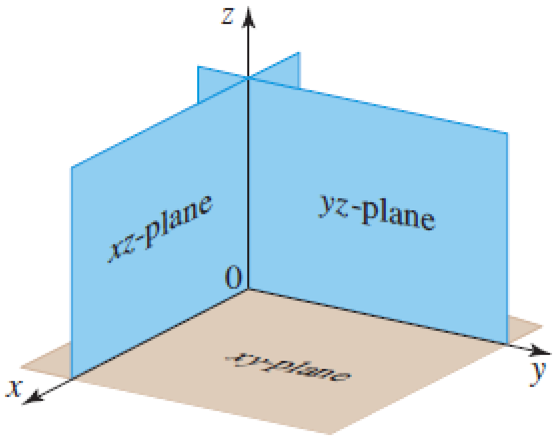
Session outline

Learning objectives

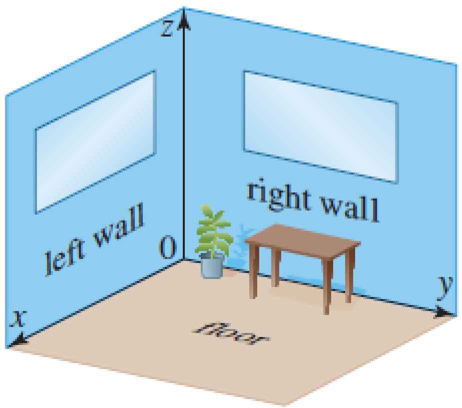
- Learn the basics of vectors in three-dimensional space, including their representation and components
- Apply the distance formula to find the distance between two points in three-dimensional space
- Conduct operations with vectors in 3D, such as addition, subtraction, and scalar multiplication
- Calculate the dot product and cross product of vectors in three dimensions and understand their geometric interpretations
- Understand and derive the equations of lines and planes in three-dimensional space, including parametric and vector equations

Vectors in 3D

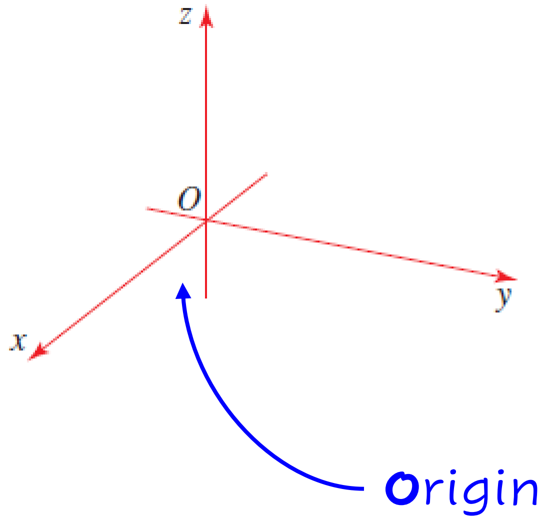
3D rectangular coordinate system



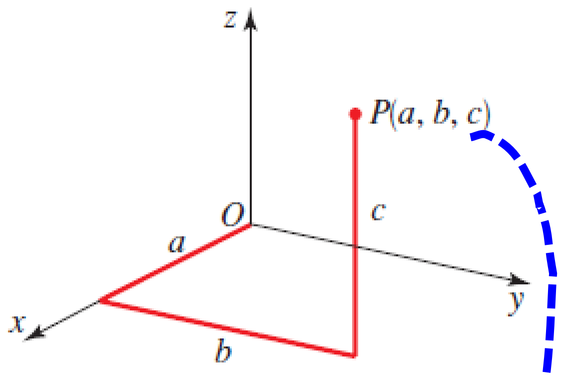
(a) Coordinate planes

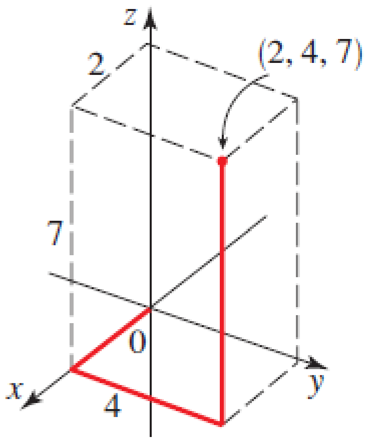


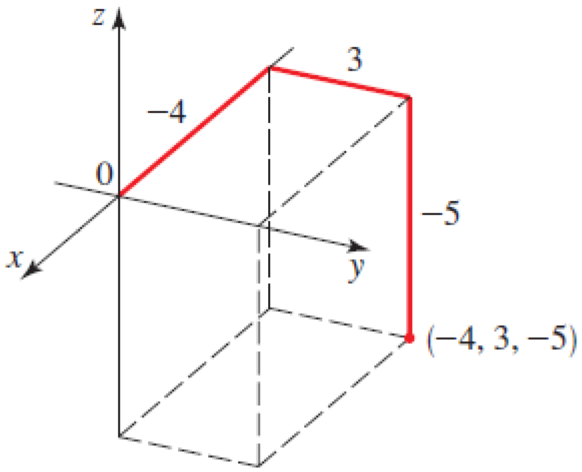
(b) Coordinate “walls”



Plotting points in three dimensions: Examples



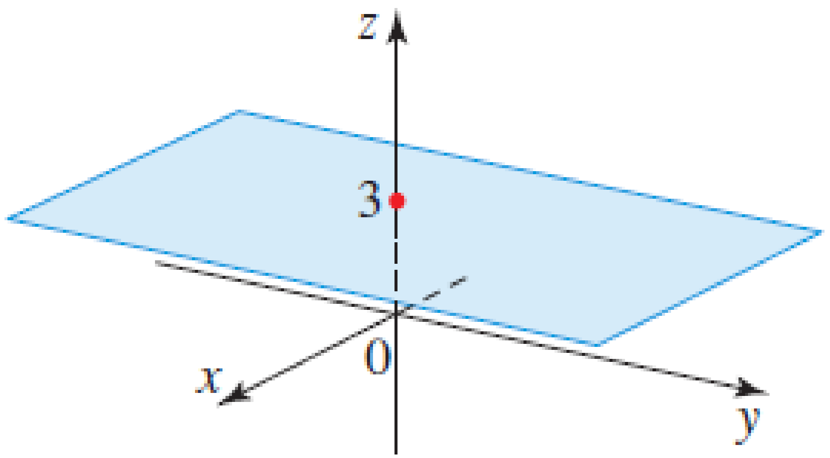




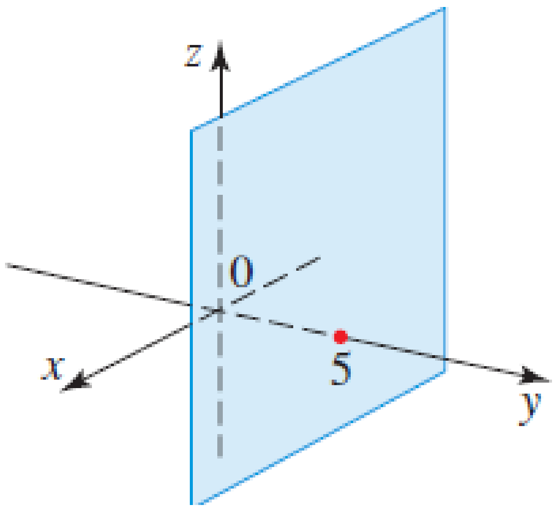
Point $P(a, b, c)$

a is the x-coordinate of P,
b is the y-coordinate of P,
c is the z-coordinate of P.

Surfaces in three-dimensional space



The plane $z = 3$



The plane $y=5$

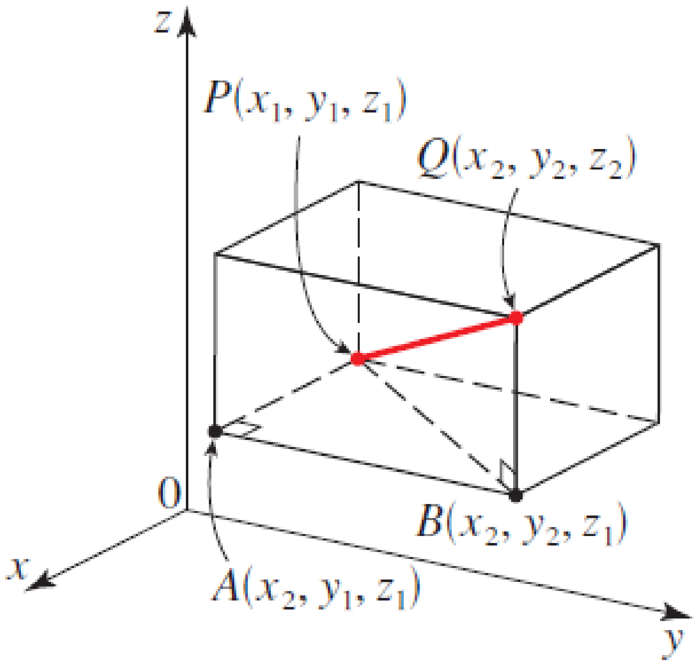
Distance formula in three dimensions

How do we measure distances in space?

DISTANCE FORMULA IN THREE DIMENSIONS

The distance between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



Using the distance formula

Find the distance between the points $P(2, -1, 7)$ and $Q(1, -3, 5)$.

SOLUTION We use the Distance Formula:

$$d(P, Q) = \sqrt{(1-2)^2 + (-3-(-1))^2 + (5-7)^2} = \sqrt{1+4+4} = 3$$

Distance in 3D

Two points *P* and *Q* are given. Find the distance between *P* and *Q*.

(a) $P(5, 0, 10), \quad Q(3, -6, 7)$

(b) $P(5, -4, -6), \quad Q(8, -7, 4)$

ACTIVITY1.1

1.1 Distance in 3D

SOLUTION

Vectors in 3D space

vectors in the coordinate plane, where the direction is restricted to two dimensions.

Vectors in space have a direction that is in three-dimensional space.

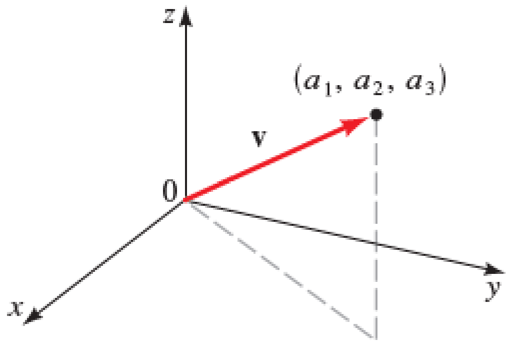


FIGURE 1 $\mathbf{v} = \langle a_1, a_2, a_3 \rangle$

COMPONENT FORM OF A VECTOR IN SPACE

If a vector $\mathbf{v}$ is represented in space with initial point $P(x_1, y_1, z_1)$ and terminal point $Q(x_2, y_2, z_2)$, then

$$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Describing vectors in component form

(a) Find the components of the vector $\mathbf{v}$ with initial point $P(1, -4, 5)$ and terminal point $Q(3, 1, -1)$.

(b) If the vector $\mathbf{w} = \langle -2, 1, 3 \rangle$ has initial point $(2, 1, -1)$, what is its terminal point?

SOLUTION

(a) The desired vector is

$$\mathbf{v} = \langle 3 - 1, 1 - (-4), -1 - 5 \rangle = \langle 2, 5, -6 \rangle$$

See Figure 2.

(b) Let the terminal point of $\mathbf{w}$ be (x, y, z). Then

$$\mathbf{w} = \langle x - 2, y - 1, z - (-1) \rangle$$

Since $\mathbf{w} = \langle -2, 1, 3 \rangle$, we have $x - 2 = -2$, $y - 1 = 1$, and $z + 1 = 3$. So $x = 0$, $y = 2$, and $z = 2$, and the terminal point is $(0, 2, 2)$.

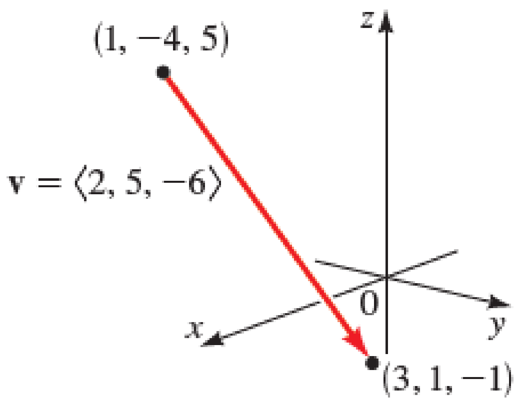


FIGURE 2 $\mathbf{v} = \langle 2, 5, -6 \rangle$

Terminal point of a vector

If the vector $\mathbf{v}$ has initial point P, what is its terminal point?

(a) $\mathbf{v} = \langle 0, 0, 1 \rangle, \quad P(0, 1, -1)$

(b) $\mathbf{v} = \langle 23, -5, 12 \rangle, \quad P(-6, 4, 2)$

1.2 Terminal point of a vector

SOLUTION

Magnitude of vectors in three dimensions

The following formula is a consequence of the Distance Formula, since the vector $\mathbf{v} = \langle a_1, a_2, a_3 \rangle$ in standard position has initial point $(0, 0, 0)$ and terminal point (a_1, a_2, a_3).

MAGNITUDE OF A VECTOR IN THREE DIMENSIONS

The magnitude of the vector $\mathbf{v} = \langle a_1, a_2, a_3 \rangle$ is

$$|\mathbf{v}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Find the magnitude of the given vector.

(a) $\mathbf{u} = \langle 3, 2, 5 \rangle$ **(b)** $\mathbf{v} = \langle 0, 3, -1 \rangle$ **(c)** $\mathbf{w} = \langle 0, 0, -1 \rangle$

SOLUTION

(a) $|\mathbf{u}| = \sqrt{3^2 + 2^2 + 5^2} = \sqrt{38}$

(b) $|\mathbf{v}| = \sqrt{0^2 + 3^2 + (-1)^2} = \sqrt{10}$

(c) $|\mathbf{w}| = \sqrt{0^2 + 0^2 + (-1)^2} = 1$

Combining vectors in space

ALGEBRAIC OPERATIONS ON VECTORS IN THREE DIMENSIONS

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$, and c is a scalar, then

$$\mathbf{u} + \mathbf{v} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

$$\mathbf{u} - \mathbf{v} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

$$c\mathbf{u} = \langle ca_1, ca_2, ca_3 \rangle$$

Operations with Three-Dimensional Vectors

If $\mathbf{u} = \langle 1, -2, 4 \rangle$ and $\mathbf{v} = \langle 6, -1, 1 \rangle$ find $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, and $5\mathbf{u} - 3\mathbf{v}$.

SOLUTION Using the definitions of algebraic operations, we have

$$\mathbf{u} + \mathbf{v} = \langle 1 + 6, -2 - 1, 4 + 1 \rangle = \langle 7, -3, 5 \rangle$$

$$\mathbf{u} - \mathbf{v} = \langle 1 - 6, -2 - (-1), 4 - 1 \rangle = \langle -5, -1, 3 \rangle$$

$$5\mathbf{u} - 3\mathbf{v} = 5\langle 1, -2, 4 \rangle - 3\langle 6, -1, 1 \rangle = \langle 5, -10, 20 \rangle - \langle 18, -3, 3 \rangle = \langle -13, -7, 17 \rangle$$

Operations with vectors

Operations with Vectors Two vectors $\mathbf{u}$ and $\mathbf{v}$ are given. Express the vector $-2\,\mathbf{u} + 3\,\mathbf{v}$ **(a)** in component form $\langle a_1, a_2, a_3 \rangle$ and **(b)** in terms of the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

(a) $\mathbf{u} = \langle 0, -2, 1 \rangle, \quad \mathbf{v} = \langle 1, -1, 0 \rangle$

(b) $\mathbf{u} = \langle 3, 1, 0 \rangle, \quad \mathbf{v} = \langle 3, 0, -5 \rangle$

ACTIVITY1.3

1.3 Operations with vectors

SOLUTION

Dot product in three dimensions

DEFINITION OF THE DOT PRODUCT FOR VECTORS IN THREE DIMENSIONS

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are vectors in three dimensions, then their **dot product** is defined by

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Calculating Dot Products for Vectors in Three Dimensions

Find the given dot product.

(a) $\langle -1, 2, 3 \rangle \cdot \langle 6, 5, -1 \rangle$

(b) $(2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) \cdot (-\mathbf{i} + 2\mathbf{j} + 8\mathbf{k})$

SOLUTION

(a) $\langle -1, 2, 3 \rangle \cdot \langle 6, 5, -1 \rangle = (-1)(6) + (2)(5) + (3)(-1) = 1$

(b) $$\begin{aligned}(2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) \cdot (-\mathbf{i} + 2\mathbf{j} + 8\mathbf{k}) &= \langle 2, -3, -1 \rangle \cdot \langle -1, 2, 8 \rangle \\ &= (2)(-1) + (-3)(2) + (-1)(8) = -16\end{aligned}$$

Dot products

Two vectors **u** and **v** are given. Find their dot product $\mathbf{u} \cdot \mathbf{v}$.

(a) $\mathbf{u} = 6\,\mathbf{i} - 4\,\mathbf{j} - 2\,\mathbf{k}, \quad \mathbf{v} = \frac{5}{6}\mathbf{i} + \frac{3}{2}\mathbf{j} - \mathbf{k}$

(b) $\mathbf{u} = 3\,\mathbf{j} - 2\,\mathbf{k}, \quad \mathbf{v} = \frac{5}{6}\mathbf{i} - \frac{5}{3}\mathbf{j}$

ACTIVITY1.4

1.4 Dot products

SOLUTION

Perpendicular vectors

Determine whether or not the given vectors are perpendicular.

(a) $4\mathbf{j} - \mathbf{k}, \quad \mathbf{i} + 2\mathbf{j} + 9\mathbf{k}$

(b) $\langle x, -2x, 3x \rangle, \quad \langle 5, 7, 3 \rangle$

ACTIVITY 1.5

1.5 Perpendicular vectors

SOLUTION

The cross product

What is cross product?

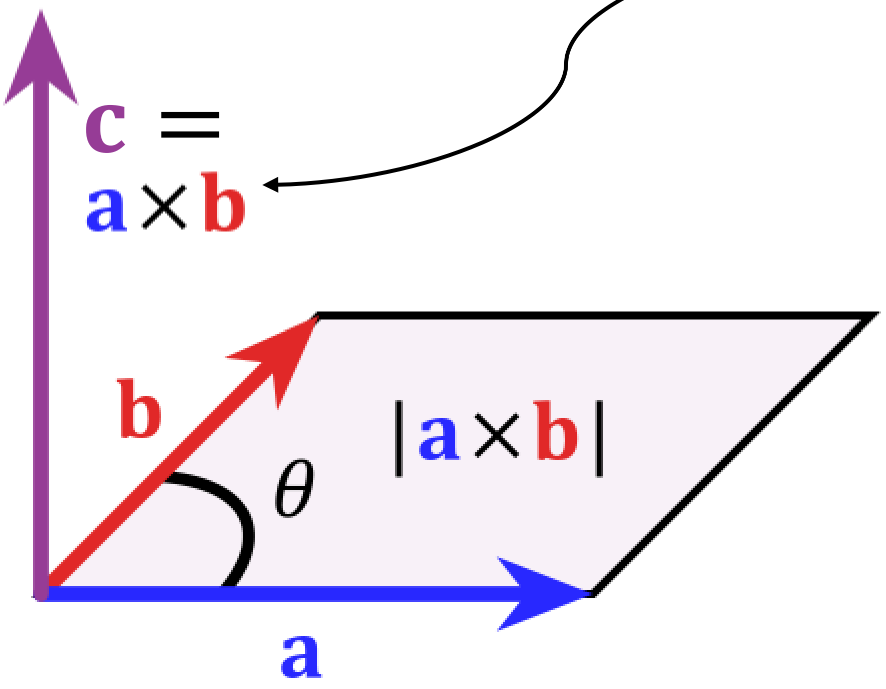
Cross product is an

operation on vectors

that allows us to **find a vector**

which is perpendicular to two given vectors.

Pronounced as
"**a cross b**"



The cross product **c** is

perpendicular to both vectors

a and **b**

$$\vec{c} \cdot \vec{a} = \vec{c} \cdot \vec{b} = 0$$

Cross product Formula

THE CROSS PRODUCT

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are three-dimensional vectors, then the **cross product** of $\mathbf{u}$ and $\mathbf{v}$ is the vector

$$\mathbf{u} \times \mathbf{v} = \langle a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1 \rangle$$

1 The ***cross product* u X v** of two vectors **u** and **v**, unlike the dot product, is a **vector** (not a scalar).

For this reason, it is also called the ***vector product***.

2 Note that the cross product is **only defined** only when **u** and **v** are vectors in **three dimensions**.

Determinants

determinant of order two

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} 2 & 1 \\ -6 & 4 \end{vmatrix} = 2(4) - 1(-6) = 14$$

determinant of order three

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ -5 & 4 & 2 \end{vmatrix} = 1 \begin{vmatrix} 0 & 1 \\ 4 & 2 \end{vmatrix} - 2 \begin{vmatrix} 3 & 1 \\ -5 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 3 & 0 \\ -5 & 4 \end{vmatrix}$$

$$= 1(0 - 4) - 2(6 - (-5)) + (-1)(12 - 0) = -38$$

Cross product using determinants

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \qquad \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \qquad \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Observe the pattern!

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$$

$$= (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}$$

Cross product using determinants

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \qquad \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \qquad \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Observe the pattern!

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$$

$$= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

Finding a cross product

If $\mathbf{u} = \langle 0, -1, 3 \rangle$ and $\mathbf{v} = \langle 2, 0, -1 \rangle$, find $\mathbf{u} \times \mathbf{v}$.

SOLUTION We use the formula above to find the cross product of $\mathbf{u}$ and $\mathbf{v}$:

$$\begin{aligned}
\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 3 \\ 2 & 0 & -1 \end{vmatrix} \\
&= \begin{vmatrix} -1 & 3 \\ 0 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 3 \\ 2 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} \mathbf{k} \\
&= (1 - 0)\mathbf{i} - (0 - 6)\mathbf{j} + (0 - (-2))\mathbf{k} \\
&= \mathbf{i} + 6\mathbf{j} + 2\mathbf{k}
\end{aligned}$$

So the desired vector is $\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$.

EXAMPLE

Finding cross products

For the given vectors **u** and **v**, find the cross product **u** X **v**.

(a) $\mathbf{u} = \langle 0, -4, 1 \rangle, \quad \mathbf{v} = \langle 1, 1, -2 \rangle$

(b) $\mathbf{u} = \langle -2, 3, 4 \rangle, \quad \mathbf{v} = \langle \frac{1}{6}, -\frac{1}{4}, -\frac{1}{3} \rangle$

ACTIVITY 2.1

2.1a Finding cross products

SOLUTION

2.1b Finding cross products

SOLUTION

Cross product theorem

CROSS PRODUCT THEOREM

The vector $\mathbf{u} \times \mathbf{v}$ is orthogonal (perpendicular) to both $\mathbf{u}$ and $\mathbf{v}$.

Proof To show that $\mathbf{u} \times \mathbf{v}$ is orthogonal to $\mathbf{u}$, we compute their dot product and show that it is 0.

$$
\begin{aligned}
(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} a_1 - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} a_2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} a_3 \\
&= a_1(a_2b_3 - a_3b_2) - a_2(a_1b_3 - a_3b_1) + a_3(a_1b_2 - a_2b_1) \\
&= \cancel{a_1a_2b_3} - \cancel{a_1a_3b_2} - \cancel{a_1a_2b_3} + \cancel{a_2a_3b_1} + \cancel{a_1a_3b_2} - \cancel{a_2a_3b_1} \\
&= 0
\end{aligned}
$$

Finding a UNIT VECTOR that is orthogonal vector to given vectors

If $\mathbf{u} = -\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} - \mathbf{k}$, find a unit vector that is orthogonal to the plane containing the vectors $\mathbf{u}$ and $\mathbf{v}$.

SOLUTION By the Cross Product Theorem the vector $\mathbf{u} \times \mathbf{v}$ is orthogonal to the plane containing the vectors $\mathbf{u}$ and $\mathbf{v}$. (See Figure 1.) In Example 1 we found $\mathbf{u} \times \mathbf{v} = \mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$. To obtain an orthogonal unit vector, we multiply $\mathbf{u} \times \mathbf{v}$ by the scalar $1/|\mathbf{u} \times \mathbf{v}|$:

$$\frac{\mathbf{u} \times \mathbf{v}}{|\mathbf{u} \times \mathbf{v}|} = \frac{\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}}{\sqrt{1^2 + 6^2 + 2^2}} = \frac{\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}}{\sqrt{41}}$$

So the desired vector is $\dfrac{1}{\sqrt{41}}(\mathbf{i} + 6\mathbf{j} + 2\mathbf{k})$.

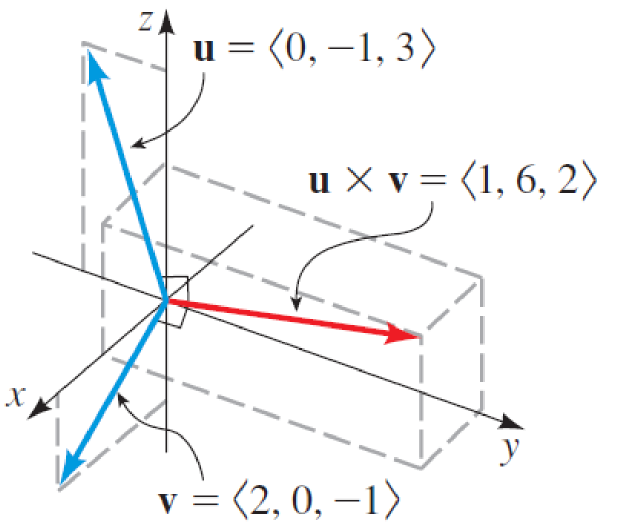


FIGURE 1 The vector $\mathbf{u} \times \mathbf{v}$ is perpendicular to $\mathbf{u}$ and $\mathbf{v}$.

EXAMPLE

Orthogonal vectors

Two vectors **u** and **v** are given.
(a) Find a vector orthogonal (perpendicular) to both **u** and **v**.
(b) Find a unit vector orthogonal (perpendicular) to both **u** and **v**.

(a) $\mathbf{u} = \frac{1}{2}\mathbf{i} - \mathbf{j} + \frac{2}{3}\mathbf{k}, \quad \mathbf{v} = 6\mathbf{i} - 12\mathbf{j} - 6\mathbf{k}$

(b) $\mathbf{u} = 3\mathbf{j} + 5\mathbf{k}, \quad \mathbf{v} = -\mathbf{i} + 2\mathbf{k}$

ACTIVITY 2.2

2.2 Orthogonal vectors

SOLUTION

2.2 Orthogonal vectors

SOLUTION

Finding a vector perpendicular to a plane

Find a vector perpendicular to the plane that passes through the points $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

SOLUTION By the Cross Product Theorem the vector $\overrightarrow{PQ} \times \overrightarrow{PR}$ is perpendicular to both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ and is therefore perpendicular to the plane through P, Q, and R. We know that

$$\overrightarrow{PQ} = (-2 - 1)\mathbf{i} + (5 - 4)\mathbf{j} + (-1 - 6)\mathbf{k} = -3\,\mathbf{i} + \mathbf{j} - 7\,\mathbf{k}$$

$$\overrightarrow{PR} = (1 - 1)\mathbf{i} + (-1 - 4)\mathbf{j} + (1 - 6)\mathbf{k} = -5\,\mathbf{j} - 5\,\mathbf{k}$$

We compute the cross product of these vectors:

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix}$$

$$= (-5 - 35)\mathbf{i} - (15 - 0)\mathbf{j} + (15 - 0)\mathbf{k} = -40\,\mathbf{i} - 15\,\mathbf{j} + 15\,\mathbf{k}$$

So the vector $\langle -40, -15, 15 \rangle$ is perpendicular to the given plane. Notice that any nonzero scalar multiple of this vector, such as $\langle -8, -3, 3 \rangle$, is also perpendicular to the plane.

Vector perpendicular to a plane

Find a vector that is perpendicular to the plane passing through the three given points.

(a) $P(0, 1, 0),\ Q(1, 2, -1),\ R(-2, 1, 0)$

(b) $P(3, 0, 0),\ Q(0, 2, -5),\ R(-2, 0, 6)$

ACTIVITY 2.3

2.3 Vector perpendicular to a plane

SOLUTION

Length of the cross product

LENGTH OF THE CROSS PRODUCT

If θ is the angle between $\mathbf{u}$ and $\mathbf{v}$ (so $0 \leq \theta \leq \pi$), then

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin \theta$$

In particular, two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if

$$\mathbf{u} \times \mathbf{v} = \mathbf{0}$$

Length of cross product

The lengths of two vectors **u** and **v** and the angle θ between them are given. Find the length of their cross product, $|u \times v|$

(a) $|\mathbf{u}| = 4, \quad |\mathbf{v}| = 5, \quad \theta = 30°$

(b) $|\mathbf{u}| = 0.12, \quad |\mathbf{v}| = 1.25, \quad \theta = 75°$

ACTIVITY 2.4

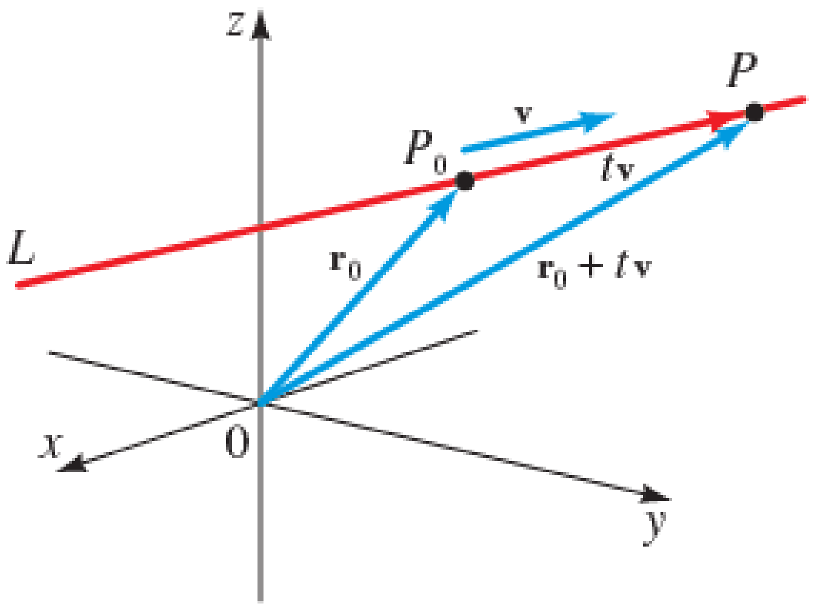
2.4 Length of cross product

SOLUTION

Equations of lines and planes

Vector equation of a line

The **position vector** of a point (a_1, a_2, a_3) is the vector $\langle a_1, a_2, a_3 \rangle$; that is, it is the vector from the origin to the point.



A line L in three-dimensional space is determined when we know a point $P_0(x_0, y_0, z_0)$ on L and the direction of L. In three dimensions the direction of a line is described by a vector $\mathbf{v}$ parallel to L. If we let $\mathbf{r}_0$ be the position vector of P_0 (that is, the vector $\overrightarrow{OP_0}$), then for all real numbers t the terminal points P of the position vectors $\mathbf{r}_0 + t\mathbf{v}$ trace out a line parallel to $\mathbf{v}$ and passing through P_0 (see Figure 1). Each value of the parameter t gives a point P on L. So the line L is given by the position vector $\mathbf{r}$, where

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

for $t \in \mathbb{R}$. This is the **vector equation of a line.**

Let's write the vector $\mathbf{v}$ in component form $\mathbf{v} = \langle a, b, c \rangle$ and let $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{r} = \langle x, y, z \rangle$. Then the vector equation of the line becomes

$$\begin{aligned} \langle x, y, z \rangle &= \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle \\ &= \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle \end{aligned}$$

Since two vectors are equal if and only if their corresponding components are equal, we have the following result.

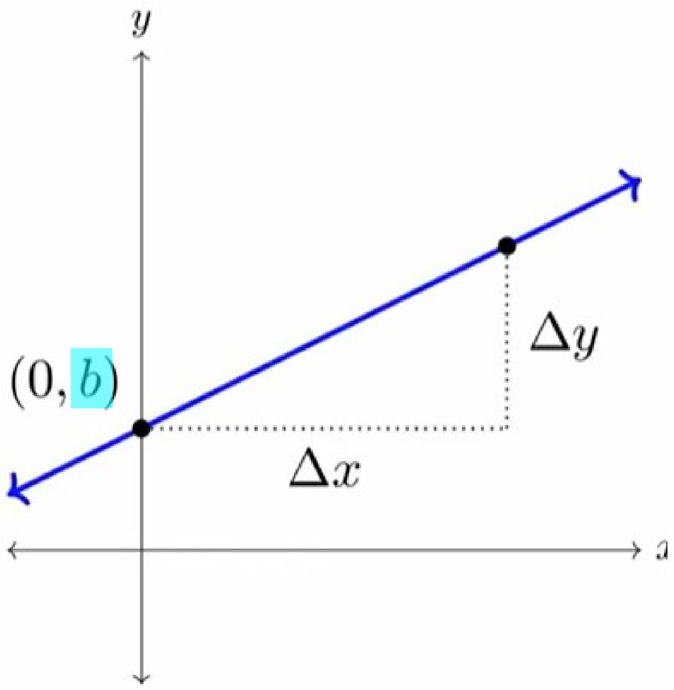
Equation of a line

Direction

Starting point

Slope-intercept form of a line: $y = mx + b$

$$m = \frac{\Delta y}{\Delta x}$$



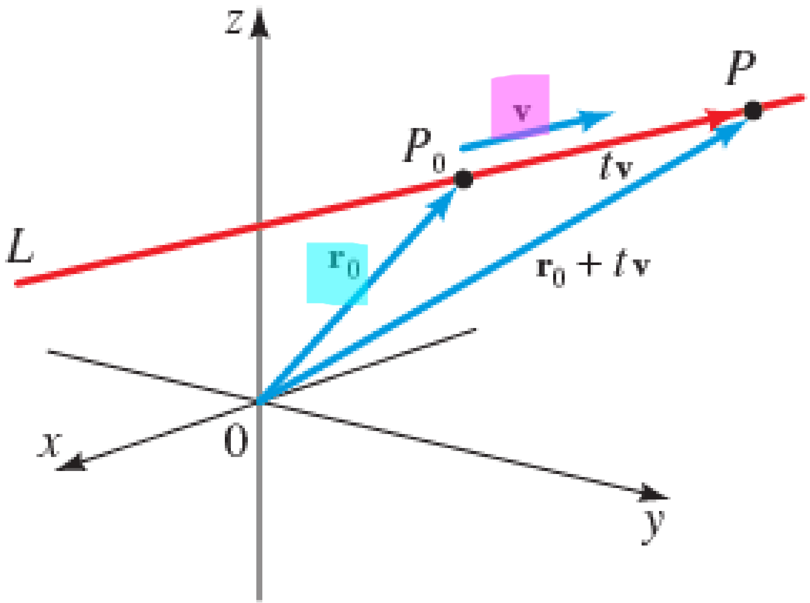
Vector Equation of a line

Starting point

Direction

$$\mathbf{r} = \mathbf{r}_0 + t\,\mathbf{v}$$

Scale factor



v is the direction vector

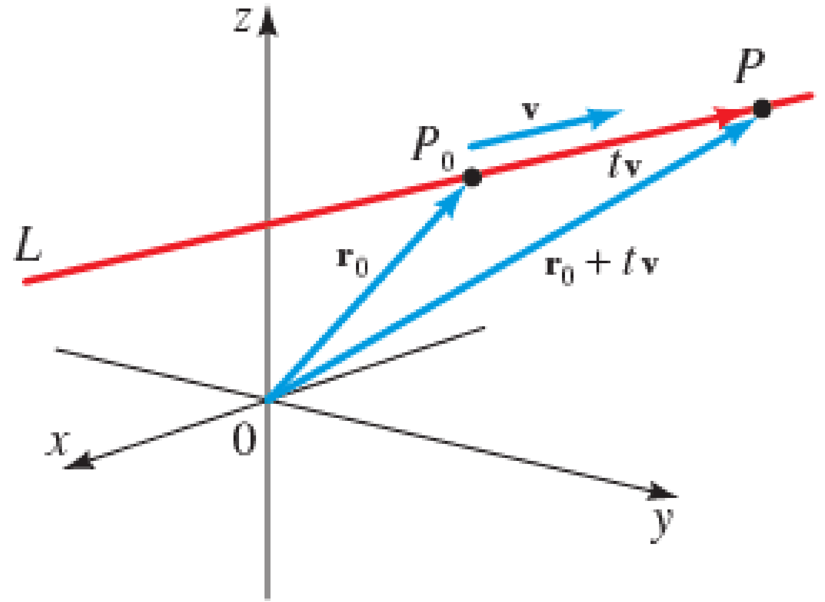
$\boldsymbol{r}_0$ is the vector from the origin to P

Vector equations of lines are not unique!

There are **multiple representations of the vector equations of a line**

- we can change the point to any other point on the line or
- we can change the length of v

So the equations of lines are **not unique**



What is a parametric equation?

A *parametric* equation is one where the x and y coordinates of a curve are **both written as functions of another variable called a parameter**;

this is usually given the letter $\boldsymbol{t}$ or $\boldsymbol{\theta}$.

Examples

$$x = \boldsymbol{t}^2$$
$$y = 2\boldsymbol{t}$$

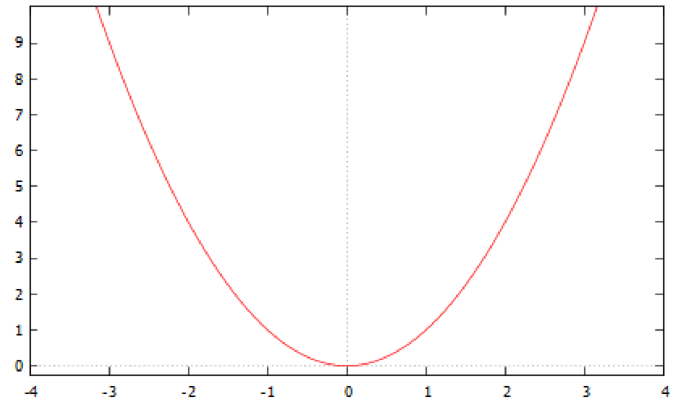
$\boldsymbol{t}$ is the **parameter**

$$x = \sin 2\boldsymbol{\theta}$$
$$y = \cos 2\boldsymbol{\theta}$$

$\boldsymbol{\theta}$ is the **parameter**

If we know '**t**' we can compute the coordinates x and y for each value of **t** and compute the curve

t	-3	-2	-1	0	1	2	3
x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9



Parametric equations for a line

PARAMETRIC EQUATIONS FOR A LINE

A line passing through the point $P(x_0, y_0, z_0)$ and parallel to the vector $\mathbf{v} = \langle a, b, c\rangle$ is described by the parametric equations

$$x = x_0 + at$$

$$y = y_0 + bt$$

$$z = z_0 + ct$$

where t is any real number.

Helps in providing information about the individual components

EXAMPLE

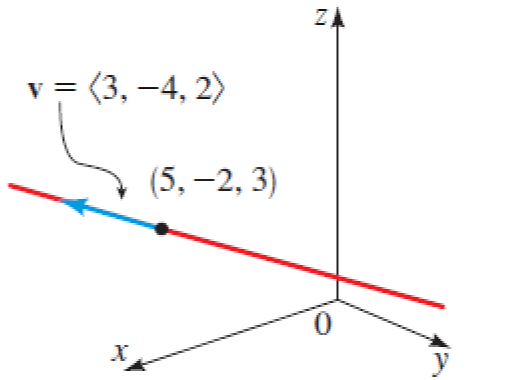


FIGURE 2 Line through $(5, -2, 3)$ with direction $\mathbf{v} = \langle 3, -4, 2\rangle$

EXAMPLE 1 ■ Equations of a Line

Find parametric equations for the line that passes through the point $(5, -2, 3)$ and is parallel to the vector $\mathbf{v} = \langle 3, -4, 2\rangle$.

SOLUTION We use the above formula to find the parametric equations:

$$x = 5 + 3t$$

$$y = -2 - 4t$$

$$z = 3 + 2t$$

where t is any real number. (See Figure 2.)

Equation of a line: Two points given

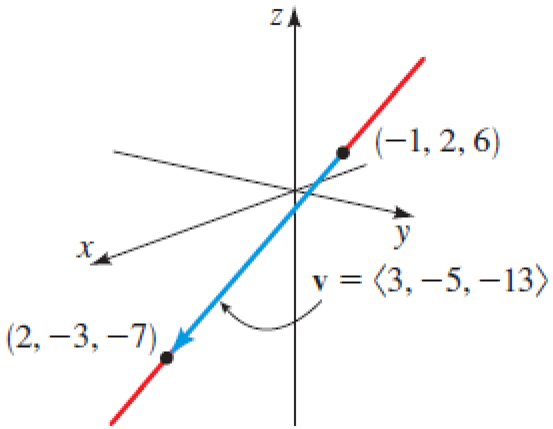


FIGURE 3 Line through $(-1, 2, 6)$ and $(2, -3, -7)$

Find parametric equations for the line that passes through the points $(-1, 2, 6)$ and $(2, -3, -7)$.

SOLUTION We first find a vector determined by the two points:

$$\mathbf{v} = \langle 2 - (-1), -3 - 2, -7 - 6 \rangle = \langle 3, -5, -13 \rangle$$

Now we use $\mathbf{v}$ and the point $(-1, 2, 6)$ to find the parametric equations:

$$x = -1 + 3t$$
$$y = 2 - 5t$$
$$z = 6 - 13t$$

where t is any real number. A graph of the line is shown in Figure 3.

EXAMPLE

Equation of a line (One point and parallel vector given)

Find parametric equations for the line that passes through the point P and is parallel to the vector $\mathbf{v}$.

(a) $P(0, -5, 3), \quad \mathbf{v} = \langle 2, 0, -4 \rangle$

(b) $P(1, 1, 1), \quad \mathbf{v} = \mathbf{i} - \mathbf{j} + \mathbf{k}$

ACTIVITY 3.1

3.1a Equation of a line (One point and parallel vector given)

SOLUTION

3.1b Equation of a line (One point and parallel vector given)

SOLUTION

Equation of lines (Two points given)

Find parametric equations for the line that passes through the points P and Q.

(a) $P(3, 3, 3)$, $Q(7, 0, 0)$

(b) $P(12, 16, 18)$, $Q(12, -6, 0)$

ACTIVITY 3.2

3.2a Equation of lines (Two points given)

SOLUTION

3.2b Equation of lines (Two points given)

SOLUTION

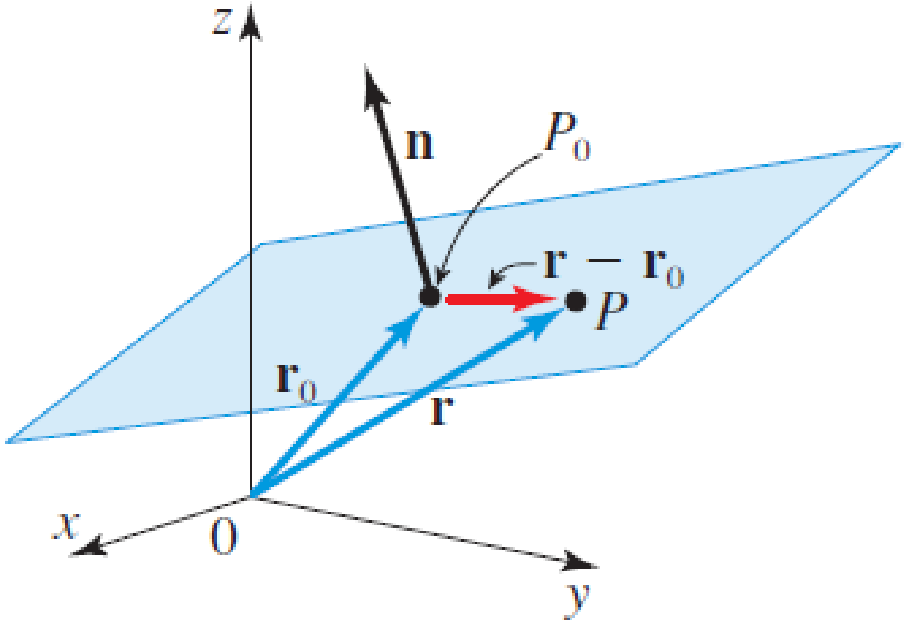
Equation of a plane

A plane is a set of terminal points of vectors
emanating from a given point P_0
perpendicular to a fixed vector $\vec{n}$

EQUATION OF A PLANE

The plane containing the point $P(x_0, y_0, z_0)$ and having the normal vector $\mathbf{n} = \langle a, b, c \rangle$ is described by the equation

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$



How did we get here?

$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$ Dot product of orthogonal vectors is zero

$\mathbf{n} = \langle a, b, c \rangle$ Normal vector

$\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ Specific point

$\mathbf{r} = \langle x, y, z \rangle$ Generic point

$$\langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Finding an equation for a plane (Normal and point given)

A plane has normal vector $\mathbf{n} = \langle 4, -6, 3 \rangle$ and passes through the point $P(3, -1, -2)$.

(a) Find an equation of the plane.

(b) Find the intercepts, and sketch a graph of the plane.

SOLUTION

(a) By the above formula for the equation of a plane we have

$$
\begin{aligned}
4(x - 3) - 6(y - (-1)) + 3(z - (-2)) &= 0 && \text{Formula} \\
4x - 12 - 6y - 6 + 3z + 6 &= 0 && \text{Expand} \\
4x - 6y + 3z &= 12 && \text{Simplify}
\end{aligned}
$$

Thus an equation of the plane is $4x - 6y + 3z = 12$.

(b) To find the x-intercept, we set $y = 0$ and $z = 0$ in the equation of the plane and solve for x. Similarly, we find the y- and z-intercepts.

x-intercept: Setting $y = 0$, $z = 0$, we get $x = 3$.

y-intercept: Setting $x = 0$, $z = 0$, we get $y = -2$.

z-intercept: Setting $x = 0$, $y = 0$, we get $z = 4$.

So the graph of the plane intersects the coordinate axes at the points $(3, 0, 0)$, $(0, -2, 0)$, and $(0, 0, 4)$. This enables us to sketch the portion of the plane shown in Figure 5.

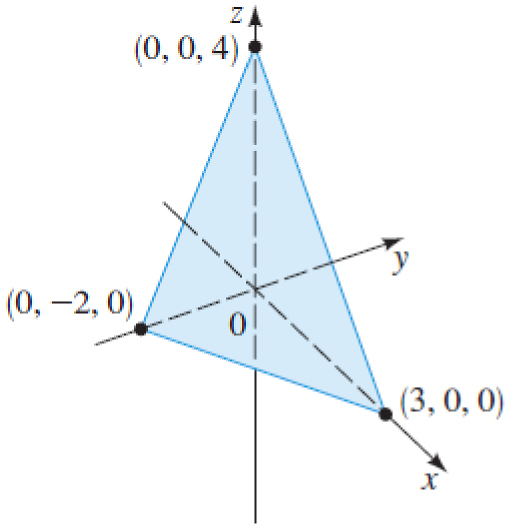


FIGURE 5 The plane $4x - 6y + 3z = 32$

Notice that in Figure 5 the axes have been rotated so that we get a better view.

EXAMPLE

Finding an equation for a plane (3 points given)

Find an equation of the plane that passes through the points $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

SOLUTION The vector $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$ is perpendicular to both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ and is therefore perpendicular to the plane through P, Q, and R. In Example 3 of Section 9.5 we found $\overrightarrow{PQ} \times \overrightarrow{PR} = \langle -40, -15, 15 \rangle$. Using the formula for an equation of a plane, we have

$$
\begin{aligned}
-40(x-1) - 15(y-4) + 15(z-6) &= 0 && \text{Formula} \\
-40x + 40 - 15y + 60 + 15z - 90 &= 0 && \text{Expand} \\
-40x - 15y + 15z &= -10 && \text{Simplify} \\
8x + 3y - 3z &= 2 && \text{Divide by } -5
\end{aligned}
$$

So an equation of the plane is $8x + 3y - 3z = 2$. A graph of this plane is shown in Figure 6.

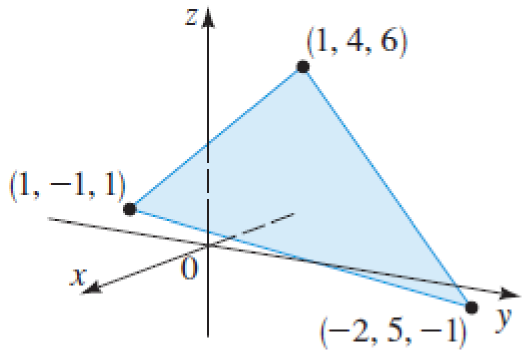


FIGURE 6 A plane through three points

EXAMPLE

equations of plane (normal vector and a point given)

A plane has normal vector **n** and passes through the point P. Find an equation for the plane.

(a) $\mathbf{n} = \langle 1, 1, -1 \rangle, \quad P(0, 2, -3)$

(b) $\mathbf{n} = \langle 3, 0, -\frac{1}{2} \rangle, \quad P(2, 4, 8)$

(c) $\mathbf{n} = \mathbf{i} + 4\mathbf{j}, \quad P(1, 0, -9)$

ACTIVITY 3.3

3.3a Equations of plane (normal vector and a point given)

SOLUTION

3.3b Equations of plane (normal vector and a point given)

SOLUTION

3.3c Equations of plane (normal vector and a point given)

SOLUTION

Equations of plane (Three points given)

Find an equation of the plane that passes through the points P, Q, and R.

(a) $P(3, 4, 5)$, $Q(1, 2, 3)$, $R(4, 7, 6)$

(b) $P(\frac{3}{2}, 4, -2)$, $Q(-\frac{1}{2}, 2, 0)$, $R(-\frac{1}{2}, 0, 2)$

ACTIVITY 2.4

3.4a Equations of plane (Three points given)

SOLUTION

3.4b Equations of plane (Three points given)

SOLUTION